

Yan Xie

yanxieyx@umich.edu | <https://yxie1010.github.io/yan-website/>

◆ Education

University of Michigan, Ann Arbor

Sep 2019 – Present

Ph.D. (degree conferral in May 2025), Climate and Space Sciences and Engineering (GPA: 4.0 / 4.0)

McGill University

Jul – Oct 2018

Undergraduate Research Trainee

Nanjing University

Sep 2015 – June 2019

Bachelor's Degree in Atmospheric Science, Graduate with honors (GPA: 3.8 / 4.0)

◆ Publications

Xie, Y., King, F., Pettersen, C., & Flanner, M. (2025). Machine Learning detection of melting layers from radar observations. *Journal of Geophysical Research: Machine Learning and Computation*, doi: [10.1029/2024JH000521](https://doi.org/10.1029/2024JH000521) [in press]

Xie, Y., Pettersen, C., Flanner, M., & Shates, J. (2024). Ground-observed snow albedo changes during rain-on-snow events in northern Alaska. *Journal of Geophysical Research: Atmospheres*, 129, doi: [10.1029/2024JD040975](https://doi.org/10.1029/2024JD040975)

Liu, Y., Huang, Y., Yuan, J., **Xie, Y.**, & Zhou, C. (2024). Contribution of surface radiative effects, heat fluxes and their interactions to land surface temperature variability. *Journal of Geophysical Research: Atmospheres*, 129, doi: [10.1029/2023JD039495](https://doi.org/10.1029/2023JD039495)

Xie, Y., Huang, X., Chen, X., L'Ecuyer, T. S., & Drouin, B. J. (2023). Joint use of far-infrared and mid-infrared observation for sounding retrievals: Learning from the past for upcoming far-infrared missions. *Earth and Space Science*, 10, doi: [10.1029/2022EA002684](https://doi.org/10.1029/2022EA002684)

Xie, Y., Huang, X., Chen, X., L'Ecuyer, T. S., Drouin, B. J., & Wang, J. (2022). Retrieval of Surface Spectral Emissivity in Polar Regions Based on the Optimal Estimation Method. *Journal of Geophysical Research: Atmospheres*, 127, doi: [10.1029/2021JD035677](https://doi.org/10.1029/2021JD035677)

L'Ecuyer, T. S., Drouin, B. J., Anheuser, J., Grames, M., Henderson, D. S., Huang, X., Kahn, B. H., Kay, J. E., Lim, B. H., Mateling, M., Merrelli, A., Miller, N. B., Padmanabhan, S., Peterson, C., Schlegel, N., White, M. L., & **Xie, Y. (2021).** The Polar Radiant Energy in the Far Infrared Experiment: A New Perspective on Polar Longwave Energy Exchanges, *Bulletin of the American Meteorological Society*, 102(7), doi: [10.1175/bams-d-20-0155.1](https://doi.org/10.1175/bams-d-20-0155.1)

Huang, Y., Chou, G., **Xie, Y.**, & Soulard, N. (2019). Radiative Control of the Interannual Variability of Arctic Sea Ice. *Geophysical Research Letters*, 46, 9899– 9908, doi: [10.1029/2019gl084204](https://doi.org/10.1029/2019gl084204)

◆ Presentations

Xie, Y., King, F., Pettersen, C., & Flanner, M. “Melting Layer Detection from Radar Observations Using Machine Learning”. **American Geophysical Union 2024 Fall Meeting.** (H34H-02 Oral Presentation)

Song, Z., Ahn, Y., Cai, J., **Xie, Y.**, Xu, X., Singh, H. “Assessing the economic impact of wildfire focus on building losses in Texas”. **Champion Team in National Science Foundation (NSF) I-GUIDE Summer School 2024: Leveraging AI for Environmental Sustainability.** (Oral Presentation)

Xie, Y., Lai, Z., Zhang, M., Gasparik, J., Webb, H., Poland, R., Cortes, A. “Aerosol influence on ground snow properties during SAIL”. **Department of Energy (DOE) Atmospheric Radiation Measurement (ARM) Open Science Summer School 2024.** (Oral Presentation)

Xie, Y., Pettersen, C., Flanner, M., & Shates, J. “Ground-observed Influence of Rainfall on Surface Snow Albedo at North Slope of Alaska”. **American Geophysical Union 2023 Fall Meeting.** (C32C-08 eLightning Presentation)

Xie, Y., Huang, X., Chen, X., L'Ecuyer, T. S., Drouin, B. J. "On the use of far-IR radiances in satellite retrievals: how can the observations collected half century ago help us preparing for the upcoming missions". **American Geophysical Union 2022 Fall Meeting.** (A32B-03 Oral Presentation)

Xie, Y., Huang, X. and Chen., X. "Retrieval of surface spectral emissivity in the polar regions: an optimal-estimation approach". **American Geophysical Union 2020 Fall Meeting.** (A239-07 Oral Presentation)

◆ **Honor and Membership**

AGU Outstanding Student Presentation Award (OSPA)	<i>March 2025</i>
AGU Precipitation Student Award 2024	<i>January 2025</i>
Rackham Predoctoral Fellowship – University of Michigan	<i>March 2024</i>
American Geophysical Union – Precipitation Technical Committee	<i>since January 2024</i>
Michigan Geophysical Union – Organization Committee	<i>January– April 2023</i>
Graduate Society of Women Engineers – University of Michigan	<i>since July 2020</i>

◆ **Teaching and Mentorship**

Graduate Student Peer Mentor – Department of Climate and Space Sciences and Engineering	<i>Fall 2023</i>
Graduate Student Instructor – Department of Climate and Space Sciences and Engineering	
CLIMATE 105: <i>Our Changing Atmosphere</i> (class size: 93 undergraduate students)	<i>Winter 2023</i>
Complete with Certificate "Preparing Future Faculty" seminar – University of Michigan	<i>Spring 2023</i>

◆ **Research Experiences (during Ph.D. study)**

Machine learning detection of melting layers using radar observation

Jan 2024 – May 2025 Rackham Predoctoral Fellow *Dept. of CLaSP, University of Michigan*
Mentors: Dr. Fraser King, Prof. Claire Pettersen, Prof. Mark Flanner

- Developed a U-Net-style semantic segmentation neural network for automatic detection of melting layers across diverse weather conditions using radar observations from DOE ARM facilities
- Performed hyperparameter tuning and machine learning model training using TensorFlow package on High Performance Computing (HPC) systems
- Assessed the machine learning model in terms of its bias, accuracy, robustness, and interpretability

Investigation of rain-on-snow events in northern Alaska using ground-based observations

Oct 2022 – Dec 2023 Graduate Student Research Assistant *Dept. of CLaSP, University of Michigan*
Mentors: Prof. Claire Pettersen, Prof. Mark Flanner, Dr. Julia Shates

- Developed a precipitation phase identification methodology to improve detection of impactful rain-on-snow events in northern Alaska
- Evaluated the impact of rainfall on the surface snow cover dynamics using snow hydrology in Community Land Model (CLM) v5.0 and a comprehensive snow albedo model SNICAR-ADv3
- Demonstrated the atmospheric drivers and moisture sources associated with these rain-on-snow events

Satellite retrievals of atmospheric profiles and surface properties in polar regions

Sep 2019 – Sep 2022 Graduate Student Research Assistant *Dept. of CLaSP, University of Michigan*
Mentors: Prof. Xianglei Huang, Dr. Xiuhong Chen

- Developed an optimal-estimation (OE) based Bayesian framework to retrieve mid- and far-infrared surface spectral emissivity using satellite observations over Arctic and Antarctic
- Assessed the ability of the remote sensing algorithm to simultaneously retrieve atmospheric profiles and surface properties using Nimbus-4 IRIS satellite observations
- Demonstrated the influence of water vapor absorption and spectral resolution on satellite remote sensing

◆ **Professional Skills**

Python, MATLAB, R, C, Fortran, NCL, LaTeX, First Aid/CPR/AED (certified by American Red Cross)